

AWS

AWS Lambda and Amazon EC2 are both cloud computing services provided by AWS, but they are designed for different purposes.

Feature	AWS Lambda	Amazon EC2
Server Management	No server management required. AWS manages the infrastructure.	User is responsible for managing the virtual server.
Scalability	Automatically scales based on incoming requests.	Scaling must be configured manually or with Auto Scaling.
Pricing	Pay only for the time the function runs.	Pay for the EC2 instance while it is running, even if idle.
Execution Time	Maximum execution time of 15 minutes.	No execution time limit.
Best For	Event-driven applications, APIs, automation, file processing.	Long-running applications, databases, web servers, and applications requiring full control.
Performance	May experience cold starts after periods of inactivity.	Consistent performance because the server is always running.
Control	Limited control over the operating system.	Full control over the operating system and installed software.

Advantages of AWS Lambda

- No server administration.
- Automatic scaling.
- Cost-effective for low or unpredictable workloads.
- Easy integration with services such as Amazon S3, API Gateway and DynamoDB.

Drawbacks of AWS Lambda

- Cold start latency.
- Maximum execution time of 15 minutes.
- Limited operating system customization.
- Can become expensive for workloads that run continuously.

Advantages of Amazon EC2

- Full control over the operating system and software.

- Suitable for long-running applications.
- Supports custom configurations and high-performance workloads.
- No execution time limit.

Drawbacks of Amazon EC2

- Requires server management and maintenance.
- Scaling is more complex than Lambda.
- Higher costs if instances remain running while idle.

Conclusion

AWS Lambda is the best choice for **event-driven and serverless applications** where automatic scaling and low operational overhead are important. Amazon EC2 is better suited for **applications that require full server control, long-running processes, or custom software configurations**. The choice depends on the application's requirements, performance needs, and cost considerations.

Yelp

Introduction

Yelp is an online platform that helps people find and review local businesses. Customers can search for restaurants, cafés, hotels, shops, healthcare providers, and many other services. They can read reviews, give businesses a rating from one to five stars, upload photos, and share their experiences. Businesses can also create or claim their profiles to provide information such as opening hours, contact details, and services offered.

History

Yelp was founded in **2004** by **Jeremy Stoppelman** and **Russel Simmons**. The company's headquarters are located in **San Francisco, California, USA**. Over the years, Yelp has become one of the world's largest online review platforms, particularly in the United States.

How Yelp Works

Yelp allows customers to search for local businesses using keywords, categories, or locations. After visiting a business, customers can leave a review and give a star rating based on their experience. Other users can then read these reviews before deciding whether to visit the business.

The basic process is:

1. A business creates or claims its Yelp profile.
2. Customers visit the business.
3. Customers leave ratings, reviews, and photos.
4. Other users use this information to make informed decisions.

Main Features

Yelp offers a variety of useful features, including:

- Business search by category or location.
- Customer reviews and ratings.
- Photo uploads.
- Maps and directions.
- Business contact information.
- Opening hours.
- Price range indicators.
- Online reservations and appointments (for selected businesses).
- Customer questions and answers.

Advantages of Yelp

Some of the main advantages of Yelp include:

- Free to use for customers.
- Large collection of customer reviews.
- Helps users compare different businesses.
- Provides real customer experiences.
- Includes photos, ratings, maps, and contact details.
- Easy to use on both computers and mobile devices.

Disadvantages of Yelp

Although Yelp is a useful platform, it also has some limitations:

- Some reviews may be fake or biased.
- Businesses may receive unfair negative reviews.
- Review filtering may hide some genuine reviews.
- It is more popular in the United States than in many other countries.
- Businesses with low ratings may lose potential customers.

Real-World Example

A customer wants to find a good Italian restaurant in London. They search for "**Italian restaurants near me**" on Yelp. The platform displays nearby restaurants together with customer ratings, reviews, photos, prices, opening hours, and directions. The customer can compare different restaurants and choose the one with the highest ratings and best reviews

Business Model

Yelp generates revenue mainly through advertising and premium business services. While customers can use the platform for free, businesses can pay to advertise their services, improve their visibility, and attract more customers.

Conclusion

Yelp is a valuable online platform that helps customers discover local businesses through reviews and ratings. It improves decision-making by providing real customer feedback, photos, and detailed business information. Although the platform has some limitations, such as fake reviews and review filtering, it remains one of the most widely used business review websites and continues to help both customers and businesses connect effectively.

Amazon Kinesis Pipeline

Introduction

An **Amazon Kinesis Pipeline** is a real-time data processing pipeline that collects, processes, and analyses streaming data as it is generated. Unlike batch processing, where data is processed at scheduled intervals, Kinesis processes data continuously, making it suitable for applications that require immediate insights.

Pipeline Components

1. Data Sources

The pipeline begins with data generated by:

- Web applications
- Mobile applications
- IoT devices
- Application logs
- Clickstream events
- Sensors

2. Amazon Kinesis Data Streams

Kinesis collects and stores streaming data in real time. It can handle thousands of records per second and distributes them across shards for scalability.

3. AWS Lambda

Lambda automatically processes incoming data from Kinesis. It can:

- Validate records
- Filter unnecessary data
- Transform data

- Trigger additional AWS services

4. AWS Glue (Streaming ETL)

Glue performs Extract, Transform and Load (ETL) operations on streaming data. It cleans, enriches, and prepares the data for storage and analysis.

5. Amazon S3

The processed data is stored in an S3 bucket, which provides secure, durable, and cost-effective storage.

6. Amazon Redshift

Redshift loads the processed data into a data warehouse where large datasets can be queried efficiently for business intelligence and reporting.

7. Amazon QuickSight or Tableau

Business intelligence tools such as Amazon QuickSight or Tableau create dashboards, charts, and reports to help users analyze the streaming data.

Advantages

- Real-time data processing.
- Highly scalable.
- Automatic scaling.
- Low latency.
- Integrates with many AWS services.
- Suitable for IoT, monitoring, fraud detection, and clickstream analytics.

Disadvantages

- More complex to design than batch pipelines.
- Higher cost for continuous data processing.
- Requires monitoring and error handling.
- Can become expensive with very high data volumes.

Real-World Example

An online shopping website tracks customer activity in real time.

1. Customers browse products.
2. Click events are sent to **Amazon Kinesis Data Streams**.
3. **AWS Lambda** validates and formats the data.
4. **AWS Glue** enriches the data with product information.
5. The processed data is stored in **Amazon S3**.
6. **Amazon Redshift** stores the data for analytics.
7. **Amazon QuickSight** displays a live dashboard showing:
 - Most viewed products
 - Active users
 - Sales trends
 - Popular categories

Conclusion

An Amazon Kinesis Pipeline enables organizations to collect, process, and analyze streaming data in real time. By integrating services such as Kinesis, Lambda, Glue, S3, Redshift, and QuickSight, businesses can gain immediate insights, monitor events continuously, and make faster data-driven decisions. It is an ideal solution for applications requiring real-time analytics, such as fraud detection, IoT monitoring, and live business reporting.